PHYS 161 Notes

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1 Units, Physical Quantities, and Vectors

1.1 SI Units

- **SI Units** refer to the system of units used by engineers and scientists across the world.
- For PHYS 161, these will be the most commonly used units in problems.

Application	Unit	Shorthand
Time	Seconds	s
Length	Meter	m
Mass	Kilogram	kg

- Additionally, many prefixes may be used in problems, necessitating conversions.

Length	Mass	Time
1 nanometer $= 1$ nm $= 10^{-9}$ m	1 microgram $= 1\ \mu$g $= 10^{-9}$ kg	1 nanosecond$= 1$ ns $= 10^{-9}$ s
1 micrometer $= 1\ \mu$m $= 10^{-6}$ m	1 milligram $= 1$ mg $= 10^{-6}$ kg	1 microsecond $= 1\ \mu$s $= 10^{-6}$ s
1 millimeter $= 1$ mm $= 10^{-3}$ m	1 gram $= 1$ g $= 10^{-3}$ kg	1 millisecond $= 1$ ms $= 10^{-3}$ s

1.2 Unit Conversions

- In equations, quantities may only be added if they have the same dimension.
- When multiplying two quantities, the product's dimension is the product of the operands' dimensions.

$$\left(\frac{\mathrm{m}}{\mathrm{s}}\right)(\mathrm{s}) = \mathrm{m}$$

1.3 Significant Figures

- The **uncertainty** of a measurement is often indicated by the number of important digits, known as the **significant figures.**
  - All non-zero digits are significant figures.

$$525 \Rightarrow \text{three significant figures}$$

  - Meanwhile, zero digits are only significant if they are between non-zero digits, or they are trailing non-zero digits when there is a decimal point.

$$40001 \Rightarrow \text{five significant figures}$$
$$120.0 \Rightarrow \text{four significant figures}$$

- When adding quantities, the significant figures of the sum are equal to the operand with the fewest significant figures.

$$25.7 + 13 = 38.7 \approx 39 \Rightarrow \text{two significant figures}$$

- When multiplying quantities, the same rule applies to the product.

$$(13.7)(20.95) = 287.015 \approx 287 \Rightarrow \text{three significant figures}$$

- In longer physics problems with numerous equations, using the actual values in calculations and only converting the final answer to the correct number of significant figures is best practice.

2 Motion Along a Straight Line

2.1 Displacement, Time, Velocity, and Acceleration

- **Time** is measured in seconds, and usually refers to the time passed between two particular points in time.
  - Denoted t.
- **Displacement** and **distance** is measured in meters.
  - Denoted Δx and x, respectively, along the x-axis.
- **Velocity** is measured in meters per second.
  - Denoted v.
- **Acceleration** is measured in meters per second squared.
  - Denoted a.

$$\begin{aligned}
\Delta x &= x_2 - x_1 \\
v_{\text{av}} &= \frac{\Delta x}{\Delta t} \\
v &= \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t} \\
a_{\text{av}} &= \frac{\Delta v}{\Delta t} \\
a &= \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2}
\end{aligned}$$

2.2 Motion with Constant Acceleration

Definition: Kinematic Equations

With constant acceleration, the following **kinematic equations** hold true:

$$\begin{aligned}
v &= v_0 + a\Delta t \\
\Delta x &= v_0 \Delta t + \frac{1}{2} a \Delta t^2 \\
v^2 &= v_0^2 + 2a\Delta x
\end{aligned}$$

Derivation: Displacement Kinematic Equation

$$\begin{aligned}
v &= v_0 + a\Delta t \\
\int_{t_1}^{t_2} v \,\mathrm{d}t &= \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t \\
\int_{t_1}^{t_2} \frac{\mathrm{d}x}{\mathrm{d}t} \,\mathrm{d}t &= \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t \\
\int_{x_1}^{x_2} \mathrm{d}x &= \int_{t_1}^{t_2} (v_0 + a\Delta t) \,\mathrm{d}t \\
\Delta x &= v_0 \Delta t + \frac{1}{2} a \Delta t^2
\end{aligned}$$

3 Motion in Two or Three Dimensions

3.1 Vector Quantities

- Many of the definitions from one-dimensional motion hold true in two or three dimensions, albeit using vectors.

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k} \qquad r = \sqrt{x + y + z}$$
$$\vec{v} = v_x\hat{\imath} + v_y\hat{\jmath} + v_z\hat{k} \qquad v = \sqrt{v_x + v_y + v_z}$$
$$\vec{a} = a_x\hat{\imath} + a_y\hat{\jmath} + a_z\hat{k} \qquad a = \sqrt{a_x + a_y + a_z}$$

- **Vector quantities** are quantities with magnitude and direction, opposed to **scalar quantities**, which only have magnitudes.

3.2 Projectile Motion

- In **projectile motion**, an object in the air only experiences acceleration due to gravity.
- Because the same kinematic equations can be applied *per component* of the vectors, the following generalizations can be made (given that z is the vertical direction):

$$v_x = v_{0x} \qquad v_y = v_{0y} \qquad v_z = v_{0z} - g\Delta t$$
$$\Delta x = v_{0x}\Delta t \qquad \Delta y = v_{0y}\Delta t \qquad \Delta z = v_{0z}\Delta t - \frac{1}{2}g\Delta t^2$$
$$v_z^2 = v_{0z}^2 - 2g\Delta z$$

Constant g is the acceleration due to Earth's gravity, $9.81\ \mathrm{m/s^2}$. Note that y is generally the vertical direction when dealing with two dimensions.

- A common application of these formulas is finding the *air time* of a projectile. Thus, sometimes the quadratic equation may be necessary to solve for the time.
  - In this case, either the negative time or the later time will be disregarded.

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Now, we can derive a general formula for a projectile's **range** and **maximum height.**

Derivation: Range Formula

Let Δx equal R and let Δy equal 0.

$$v_y = v_{0y} - gt$$
$$gt = v_{0y} - v_y$$
$$t = \frac{v_0 \sin\theta - v\sin\theta}{g}$$
$$t = \frac{v_0 \sin\theta - (-v_0 \sin\theta)}{g}$$
$$t = \frac{2v_0 \sin\theta}{g}$$

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Derivation: Range Formula *continued*

Now, we can plug this value for t into a horizontal kinematic equation.

$$R = v_{0x}t$$
$$R = v_0 \cos\theta t$$
$$R = v_0 \cos\theta \frac{2v_0 \sin\theta}{g}$$
$$R = \frac{v_0^2 \sin(2\theta)}{g}$$

Derivation: Maximum Height Formula

Let Δy equal h_{max} and $v_y = 0$.

$$0 = v_{0y}^2 - 2gh_{\text{max}}$$
$$2gh_{\text{max}} = v_{0y}^2$$
$$h_{\text{max}} = \frac{v_{0y}^2}{2g}$$
$$h_{\text{max}} = \frac{v_0^2 \sin^2\theta}{2g}$$

- In later chapters, there will be other ways of finding the range or maximum height of a projectile, but these may still be useful depending on what is known in a problem.

Example: Fireworks Display

During a fireworks display, a shell is shot into the air with an initial speed $v_0 = 70.0$ m/s at an angle of 75° above the horizontal. The fuse is timed to ignite the shell just as it reaches its highest point above the ground.

1. How much time has passed between the launch of the shell and the explosion?
2. What is the horizontal displacement of the shell before it explodes?

---

1) This is essentially the time needed for the firework to reach its max height. We can derive the maximum height formula (as seen above) first.

$$h_{\text{max}} = \frac{(70.0 \text{ m/s})^2 \sin^2 75°}{2(9.81 \text{ m/s}^2)} = 233 \text{ m}$$

Now, we can solve for time easily.

$$h_{\text{max}} = v_{0y}t - \frac{1}{2}gt^2$$
$$h_{\text{max}} = v_0 \sin(75°)t - \frac{1}{2}gt^2$$
$$\frac{1}{2}gt^2 - v_0 \sin(75°)t + h_{\text{max}} = 0$$
$$\boldsymbol{t = 6.89 \text{ s}}$$

Continued on next page

Example: Fireworks Display *continued*

2) We can calculate the firework's displacement using the time we just calculated.

$$\begin{aligned}\Delta x &= v_{0x}t \\ \Delta x &= v_0 \cos(75°)t \\ \Delta x &= (70\ \text{m/s})\cos(75°)(6.89\ \text{s}) \\ \boldsymbol{\Delta x} &= \mathbf{125\ m}\end{aligned}$$

3.3 Circular Motion

- Even if a particle's speed is constant, its velocity cannot be constant in circular motion because *its direction is constantly changing.*
- The direction of the velocity changes due to acceleration directed toward the center of its rotation, known as **centripetal acceleration.**
  - Denoted a_t or a_{rad}.
  - Thus, when centripetal acceleration is the only acceleration vector acting on a particle, the particle undergoes **uniform circular motion** because it will always be perpendicular to the particle's velocity vector.
- An acceleration vector parallel to the particle's velocity vector is known as **tangential acceleration.**
  - Denoted a_t.
  - This is known as **nonuniform circular motion**, because the speed of the particle will also be changing.

$$a_t = \frac{\mathrm{d}v}{\mathrm{d}t}$$

- **Frequency** is the rate at which an object spins, measured in revolutions per second (rev/s) or sometimes just in hertz ($\text{s}^{-1} = \text{Hz}$).
  - Denoted f.
- **Period** is the time required to make one revolution, measured in seconds.
  - Denoted T.

$$T = \frac{1}{f}$$

Definition: Circular Motion Equations

Given radius of the circular motion R

$$v = \frac{2\pi R}{T} = 2\pi R f$$

$$a_c = \frac{v^2}{R} = \frac{4\pi^2 R}{T} = 4\pi^2 R f$$

$$a_{\text{tot}} = \left\| \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} \right\| = \sqrt{a_c^2 + a_t^2}$$

3.4 Relative Motion

- Given reference frames A, B, and C where AB means "A in the reference frame of B"

$$\vec{v}_{AB} + \vec{v}_{BC} = \vec{v}_{AC}$$
$$\vec{v}_{AB} = -\vec{v}_{BA}$$

- This is essentially vector addition.
- The **Law of Sines** and the **Law of Cosines** are useful formulas for relative motion problems whenever angles are needed.

Definition: Law of Sines

Given sides a, b, and c opposite of angles A, B, and C, respectively,

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Definition: Law of Cosines

Given sides a, b, and c opposite of angles A, B, and C, respectively,

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = a^2 + c^2 - 2ac\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

Example: Metro Trains

Two metro trains pass by each other on the red line. The northbound train is moving at a speed of 25.0 m/s and the southbound train is moving at a speed of 20.0 m/s. In the northbound train, Cecilia is walking toward the front of the train at 2.0 m/s, and in the southbound one, Jack is moving toward the back of the train also at 2.0 m/s. What is the magnitude of Jack's velocity relative to Cecilia?

We can start by finding both of their speeds relative to the Earth.

$$\vec{v}_{\text{Cecilia,Earth}} = \vec{v}_{\text{Cecilia,North}} + \vec{v}_{\text{North,Earth}}$$
$$\vec{v}_{\text{Cecilia,Earth}} = (2\ \text{m/s}) + (25\ \text{m/s}) = 27\ \text{m/s}$$
$$\vec{v}_{\text{Jack,Earth}} = \vec{v}_{\text{Jack,South}} + \vec{v}_{\text{South,Earth}}$$
$$\vec{v}_{\text{Jack,Earth}} = (2\ \text{m/s}) + (-20\ \text{m/s}) = -18\ \text{m/s}$$

Now, we can solve for Jack's velocity relative to Cecilia.

$$\vec{v}_{\text{Jack,Cecilia}} = \vec{v}_{\text{Jack,Earth}} + \vec{v}_{\text{Earth,Cecilia}}$$
$$\vec{v}_{\text{Jack,Cecilia}} = \vec{v}_{\text{Jack,Earth}} - \vec{v}_{\text{Cecilia,Earth}}$$
$$\vec{v}_{\text{Jack,Cecilia}} = (-18\ \text{m/s}) - (27\ \text{m/s})$$
$$\vec{v}_{\text{Jack,Cecilia}} = -45.0\ \text{m/s}$$
$$\boldsymbol{v}_{\textbf{Jack,Cecilia}} = \mathbf{45.0\ m/s}$$

4 Newton's Laws of Motion

4.1 Newton's First Law

Definition: Newton's First Law
An object subject to no external forces is at rest or moves with constant velocity if viewed from an inertial reference frame.

$$\text{Inertial Reference Frame} \equiv \text{Reference frame in which Newton's laws are valid}$$

- An inertial reference frame must not be accelerating or rotating.

4.2 Newton's Second Law

- **Forces** are pushes or pulls on objects, measured in Newtons (N) or $(\mathrm{kg} \cdot \mathrm{m/s}^2)$.

Definition: Newton's Second Law

$$\Sigma \vec{F} = m\vec{a}$$

4.3 Newton's Third Law

Definition: Newton's Third Law
For every action there is an equal and opposite reaction.

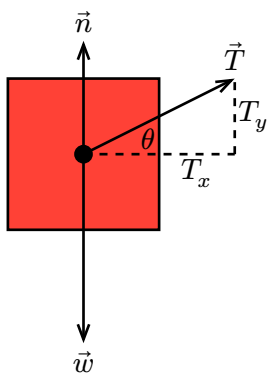
Given objects A and B where AB means "exerted from A on B"

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

- For example, when two objects collide, they experience the same magnitude of force from the collision.

4.4 Free Body Diagrams

- **Free-body diagrams** show the forces acting on an object's center, including any relevant components.
- While the magnitude of the drawn forces isn't that important, scaling them relative to each other makes the direction of the net acceleration clearer.



Free-body diagram of a block being pulled at an angle θ by a rope. Note how the vertical components of the forces are all balanced, signifying that it is being dragged horizontally.

5 Applying Newton's Laws

5.1 Types of Forces

- **Fundamental forces** are the four natural interactions that govern how particles interact.
  - **Weight**
  - Electromagnetic
  - Weak nuclear
  - Strong nuclear
- **Non-fundamental forces** are forces whose interactions are derived from the fundamental forces.
  - **Spring**
  - **Contact** (tension, normal, friction, ...)
  - Microscopic
- The **weight** force is from gravity. On Earth, it is defined as follows:

$$w = mg$$

- However, there also exists a general definition for weight force, the **Universal Law of Gravitation** using the gravitational constant G:

$$G = 6.6743 \times 10^{-11}\ \mathrm{m}^3 \cdot \mathrm{kg}^{-1} \cdot \mathrm{s}^{-1}$$

Definition: Universal Law of Gravitation

The force between any two objects is defined as follows, given that r is the *distance between the objects' centers*

$$F_1 = F_2 = G\frac{m_1 m_2}{r^2}$$

- The **spring force** is from compressing a spring.
  - Springs have differing **spring constants** usually denoted as k, and they define the ratio between the force exerted by the spring and the distance compressed from the equilibrium position.

$$\vec{F}_{\text{spr}} = -k\vec{x}$$

- The **normal force** is exerted by a surface on an object.
  - Generally denoted $\vec{n}$.
- The **tension force** is the pulling force exerted equally throughout a rope, cable, etc.
  - Generally denoted $\vec{T}$.
- The **friction force** is exerted by a surface on an object and acts perpendicular to the normal force. It depends on the roughness of the surface (expressed by the coefficient of friction).
  - **Kinetic friction**, denoted $\vec{f}_k$ with coefficient of kinetic friction μ_k, acts on objects sliding across a surface.

$$\vec{f}_k = \mu_k \vec{n}$$

  - **Static friction**, denoted $\vec{f}_s$ with coefficient of static friction μ_s, acts on still objects until the applied force exceeds the maximum static friction.

$$\vec{f}_s \leq \vec{f}_{s,\max} = \mu_s \vec{n}$$

5.2 Forces in Equilibrium

- Objects are in equilibrium when they are at rest or moving with constant velocity, meaning that the net force acting on those objects must be zero.

$$\Sigma \vec{F} = 0$$

- A common example of forces in equilibrium is an object sitting on a horizontal surface. The net vertical force is zero because the magnitude of the normal force and the weight force are equal.

5.3 Dynamics of Particles

- Using Newton's second law, we can set all forces (or components of forces) equal to mass times acceleration and solve for unknowns accordingly.
- Signs can indicate the direction, but you have to be consistent.
  - In **pulley problems**, direction should be based on the direction of the pulley's cord.
  - In **incline problems**, the x-axis should be parallel to the incline and the y-axis should be perpendicular to the incline.
    - Consequently, when given the angle of the incline, sine is used to find the x-component of weight and cosine is used to find the y-component of weight because weight will not be perpendicular to the incline.
- Sometimes, normal force can be referred to as **apparent weight** because the magnitude may be less than or greater than the opposing weight force.
  - For instance, you feel *heavier* when you are in an elevator moving up because your normal force is greater than your weight force and vice versa.

$$n = m(g + a_y)$$

Example: Table Pulley

Block B with mass $m_B = 5.0$ kg rests on block A, with mass $m_A = 8.0$ kg, which is on a horizontal frictionless tabletop. The coefficient of static friction between block A and block B is $\mu_s = 0.75$. A string attached to block A passes over a massless, frictionless pulley at the edge of the tabletop, and block C is suspended from the other end of the string and is released from rest.

1. Write Newton's second law for each of the three blocks in both directions.
2. Find the largest mass that block C can have so that blocks A and B slide together when the system is released from rest.

---

1) Along the direction of the pulley, the vertical forces acting on block A and B are irrelevant.

$$\begin{aligned} \Sigma F_A &= T - f_s \\ \Sigma F_B &= f_s \\ \Sigma F_C &= m_C g - T \end{aligned}$$

2) Now, blocks A and B will slip once the maximum static friction is exceeded, so we can use their net force equations to find the acceleration of the system when m_C is maximized.

$$\begin{aligned} m_B a &= \mu_s m_B g \\ a &= \mu_s g \\ a &= (0.75)(9.81 \text{ m/s}^2) = 7.36 \text{ m/s}^2 \end{aligned}$$

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Example: Table Pulley *continued*

Using this, we can find an expression for T.

$$\begin{aligned} T - f_s + f_s &= (m_A + m_B)a \\ T &= (m_A + m_B)a \end{aligned}$$

With both acceleration and tension found, we can now solve for the maximum mass of C.

$$\begin{aligned} m_C a &= m_C g - T \\ m_C a &= m_C g - (m_A + m_B)a \\ m_C a - m_C g &= -(m_A + m_B)a \\ m_C &= -\frac{(m_A + m_B)a}{a - g} \\ m_C &= -\frac{(8\text{ kg} + 5\text{ kg})(7.36\text{ m/s}^2)}{7.36\text{ m/s}^2 - 9.81\text{ m/s}^2} \\ \boldsymbol{m_C} &= \mathbf{39\text{ kg}} \end{aligned}$$

5.4 Dynamics of Circular Motion

- Based on Newton's Second Law, circular motion can result from many different forces.

$$\Sigma F = ma_c = \frac{mv^2}{R}$$

- As a result, denoting a singular "centrifugal force" is generally not accurate.
  - For example, if an object is moving vertically with uniform circular motion, the weight force points toward the center of rotation at the top, but away from the center of rotation at the bottom.
- Additionally, direction should be considered relative to the center of rotation—generally, the positive direction is toward the center.
- Additionally, problems involving **circular motion of vehicles** have less obvious forces contributing to the circular motion.
  - Along a curved road with friction, *static friction is what maintains the vehicle's circular motion* because at any given moment, the tires of the car are stationary relative to the pavement.
  - Along a banked curve without friction (sort of like an inclined plane from the inner curve to the outer curve), *the x-component of the normal force is what maintains the vehicle's circular motion.*

Example: Rounding a Flat Curve

A sports car is rounding a flat, unbanked curve with radius R. If the coefficient of static friction between the tires is μ_s, what is the maximum speed v_{max} the driver can complete the curve without sliding off?

Here, the only force contributing to the circular motion is static friction, so we can just use the given formula for maximum static friction to find v_{max}.

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Example: Rounding a Flat Curve *continued*

$$ma_c = f_s$$
$$\frac{mv_{\text{max}}^2}{R} = \mu_s mg$$
$$\boldsymbol{v_{\text{max}} = \sqrt{R\mu_s g}}$$

Note that the maximum speed needed is not based on the mass of the vehicle at all.

Example: Rounding a Banked Curve

You are planning to create a road banked at the perfect angle to allow cars to complete turns without friction. At what angle α should the road be banked to allow for this?

We can imagine this from a straight back perspective of the car. While similar to an inclined plane problem, we are evaluating components parallel to the horizontal rather than the incline. As a result, the only inward force is the x-component of the normal force because the curve is banked from the inside-out.

Now, because the normal force and its y-component are perpendicular to the respective sides forming the angle α, *the angle between them is also equal to* α. Thus, the x-component of the normal force is the component opposite of α while the y-component of the normal force is the component adjacent of α.

$$n_y = mg$$
$$n\cos\alpha = mg$$
$$n = \frac{mg}{\cos\alpha}$$

Using this value of n, we can now solve for the angle.

$$n_x = \frac{mv^2}{R}$$
$$n\sin\alpha = \frac{mv^2}{R}$$
$$mg\frac{\sin\alpha}{\cos\alpha} = \frac{mv^2}{R}$$
$$mg\tan\alpha = \frac{mv^2}{R}$$
$$\tan\alpha = \frac{v^2}{Rg}$$
$$\boldsymbol{\alpha = \arctan\left(\frac{v^2}{Rg}\right)}$$

6 Work and Kinetic Energy

6.1 Work

- A force does **work** on a body if the body undergoes a displacement.
- The units of work are joules (J) or newton-meters.

Definition: Work
Given work W, force $\vec{F}$, and displacement $\vec{s}$

$$W = \vec{F} \cdot \vec{s} = Fs\cos(\theta)$$

- Because work is defined by a dot product, it is a scalar quantity, although it can still be negative.
- Thus, it follows that a force does **negative work** when applied opposite of the direction of $\vec{s}$ and does **zero work** when applied perpendicular to the direction of $\vec{s}$.
- Additionally, given this definition, there are multiple ways to define **net work.**

$$W_{\text{net}} = \Sigma\vec{F} \cdot \vec{s} = \Sigma W$$

6.2 Kinetic Energy

- **Energy** is the capacity to do work, and also uses joules.
- **Kinetic energy** is the energy of a body from its mass and speed.

Definition: Kinetic Energy
Given kinetic energy K, mass m, and speed v

$$K = \frac{1}{2}mv^2$$

- Because kinetic energy is based on speed, it is either positive or zero.
- Additionally, work may be expressed as the **change in kinetic energy.**

$$W = \Delta K$$

Derivation: Work-Energy Theorem
Given work W and kinetic energy K

$$F_{\parallel} = ma = m\frac{v_2^2 - v_1^2}{2s}$$
$$F_{\parallel}s = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$
$$W = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \Delta K$$

- It follows that energy is also a scalar quantity.
- However, these formulas only hold when the force remains constant during the displacement.

Example: Final Velocity of a Sled

You push a 30.0 kg sled with constant horizontal force $F = 500$ N for a horizontal distance $d = 20.0$ m. The sled's initial velocity is 10.0 m/s. What is the final velocity of the sled?

---

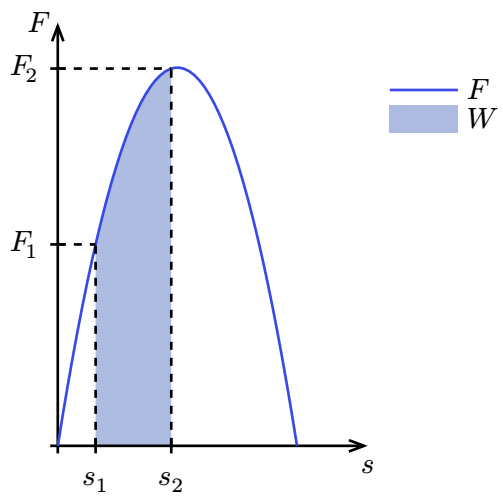
1) Because we can calculate the work done and we are given the initial velocity, we can use the work-energy theorem to derive an expression for the final velocity.

$$\begin{aligned} W &= F_{\parallel} d \\ W &= (500\ \text{N})(20\ \text{m}) \\ W &= 10000\ \text{J} \\ \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 &= 10000\ \text{J} \\ \frac{1}{2}mv_2^2 &= 10000\ \text{J} + \frac{1}{2}mv_1^2 \\ v_2 &= \sqrt{\frac{20000\ \text{J}}{m} + v_1^2} \\ v_2 &= \sqrt{\frac{20000\ \text{J}}{30.0\ \text{kg}} + (10\ \text{m/s})^2} \\ \boldsymbol{v_2} &= \mathbf{27.7\ m/s} \end{aligned}$$

6.3 Work and Energy with Varying Force

- If an object is experiencing a varying net force, then the total work can be approximated using the average force per increment of the total displacement.
- Thus, as the increment tends to zero, work may be defined as follows:

$$W = \int_{s_1}^{s_2} F\,\mathrm{d}s$$



- In more than one dimension:

$$W = \int_{r_1}^{r_2} \vec{F} \cdot \mathrm{d}\vec{r}$$

- This is also the definition for work done by a spring because it varies depending on the displacement.

Derivation: Work Done by a Spring

$$W_s = \int_{x_1}^{x_2} (-kx)\,\mathrm{d}x$$

$$W_s = -\frac{1}{2}k\Delta x^2$$

6.4 Power

- **Power** is the rate at which work is done, measured in watts (W) or joules per second (J/s).
  - Denoted P.
  - Power is also sometimes given in horsepower (hp). The conversion rate is as follows:

$$1\ \mathrm{hp} = 745.7\ \mathrm{W}$$

- Thus, average power is defined as follows:

$$P_{\mathrm{av}} = \frac{\Delta W}{\Delta t}$$

Definition: Power

Given power P, work W, velocity $\vec{v}$, and force $\vec{F}$

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \vec{F} \cdot \vec{v} = Fv\cos\theta$$

7 Potential Energy and Energy Conservation

- **Potential energy** is energy based on position, and may be *converted* into other forms of energy depending on the change in position.
  - For example, a fruit hanging from a tall tree has the *potential* to reach a higher speed (and therefore kinetic energy) before it hits the ground than a fruit hanging from a smaller tree.

7.1 Gravitational Potential Energy

- **Gravitational potential energy** is potential energy dependent on an object's mass and height above the ground.
  - Denoted U_g.

Definition: Gravitational Potential Energy
Given gravitational potential energy U_g, object's mass m, acceleration due to gravity g, and height above the ground h

$$U_g = mgh$$

- Notice how an object's gravitational potential energy *decreases* as its height above the ground decreases, even though the work done by the weight force on a falling object is *positive*.
- Thus, the work done by gravity is equal to the negative change in gravitational potential energy.

$$W_g = -\Delta U_g$$

7.2 Elastic Potential Energy

- **Elastic potential energy** is potential energy stored in deformable objects such as springs when they are stretched or compressed from equilibrium.
  - Denoted U_s.

Definition: Elastic Potential Energy
Given elastic potential energy U_s, spring constant k, and elongation of object x

$$U_s = \frac{1}{2}kx^2$$

- Similarly to work done by gravity, because an object's elastic potential energy increases the more it is displaced from equilibrium, the work done by the spring force is equal to the negative change in elastic potential energy.

$$W_s = -\Delta U_s$$

7.3 Conservative and Nonconservative forces

- **Mechanical energy** is energy that is either kinetic or potential.
- **Conservative forces** do work independent of the path, meaning that they only depend on the endpoints of the displacement.
  - The work done by conservative forces is reversible because they convert between forms of mechanical energy.
- **Nonconservative forces** do work dependent on the path.

- The work done by nonconservative forces is irreversible because they convert forms of mechanical energy to other forms of energy.

The Law of Conservation of Energy

- Thus, as long as the net work done by nonconservative forces in a system is zero, the change in total mechanical energy is zero.
- As a result, we can equate the total mechanical energy at different points in a system.
- Thus, work done by nonconservative forces should be subtracted.

Definition: Law of Conservation of Energy

Given kinetic energy K, potential energy U, and work from nonconservative forces W_{nc}

$$\Delta K + \Delta U - W_{\text{nc}} = 0$$

- For conservation of energy problems, the following form is often used:

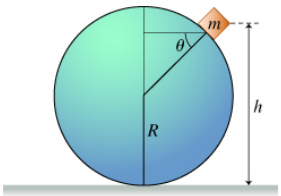
$$K_1 + U_1 + W_{\text{nc}} = K_2 + U_2$$

- Nonconservative forces may also be represented using **internal energy** U_{int} such that the internal energy changes when external forces do work on the system.

$$\Delta U_{\text{int}} = -W_{\text{nc}}$$

Example: Block on a Globe

A small block of mass m sits on top of a globe with radius R. The small block begins sliding from the top of the globe with negligible speed. The block leaves the surface of the globe when it reaches a height h_{crit} above the ground. Derive a formula for v_{crit}, the speed of the block the moment it falls off the globe, in terms of R, h_{crit}, and g.



The first step is to identify two points in the system that would include the desired terms. One of them should be when the block is falling off because it is at height h_{crit}. The other point should be at the top of the globe because the height above the ground is just the diameter of the globe.

$$\begin{aligned}
\frac{1}{2}mv_{\text{crit}}^2 + mgh_{\text{crit}} &= mg(2R) \\
\frac{1}{2}mv_{\text{crit}}^2 &= mg(2R) - mgh_{\text{crit}} \\
v_{\text{crit}}^2 &= 4gR - 2gh_{\text{crit}} \\
\boldsymbol{v_{\text{crit}}} &= \boldsymbol{\sqrt{4gR - 2gh_{\text{crit}}}}
\end{aligned}$$

Example: Throwing a Baseball

The maximum horizontal distance you can throw a 0.145 kg baseball is 25 m at an angle of $\alpha = 45°$ above the horizontal.

1. Ignoring air resistance and the small distance the baseball is above the ground when it is thrown, what is the kinetic energy of the baseball the moment it is thrown?
2. At the ball's maximum height, what fraction of its total mechanical energy is kinetic energy?

---

1) Because we are given the range of the baseball but not the initial velocity, we can use our previously derived range formula to find the initial velocity.

$$R = \frac{v_0^2 \sin(2\alpha)}{g}$$
$$v_0^2 = \frac{Rg}{\sin(2\alpha)}$$
$$v_0 = \sqrt{\frac{Rg}{\sin(2\alpha)}}$$
$$v_0 = \sqrt{\frac{(25\ \text{m})(9.81\ \text{m/s}^2)}{\sin 90°}}$$
$$v_0 = 15.7\ \text{m/s}$$

Now we can plug this back into the kinetic energy formula.

$$K_0 = \frac{1}{2}mv_0^2$$
$$K_0 = \frac{1}{2}(0.145\ \text{kg})(15.7\ \text{m/s})^2$$
$$\boldsymbol{K_0 = 17.8\ \text{J}}$$

2) Because we know that the vertical component of the velocity vector is zero at the highest point in its trajectory, the velocity at the maximum height is just the horizontal component.

$$K_1 = \frac{1}{2}mv_1^2$$
$$K_1 = \frac{1}{2}mv_0^2 \cos^2(45°)$$
$$K_1 = \frac{1}{2}mv_0^2\left(\frac{1}{2}\right)$$
$$K_1 = \frac{1}{2}K_0 = \frac{1}{2}E_{\text{tot}}$$
$$\boldsymbol{\frac{K_1}{E_{\text{tot}}} = \frac{1}{2}}$$

Example: Rough Patch

A small block is sent through point A with a speed of $7\ \mathrm{m/s}$. Point A is at height $h_1 = 6\ \mathrm{m}$. The block then slides down to point B, which is at a height of zero. After that, the block slides up to point C at a height of $h_2 = 2\ \mathrm{m}$. Past point C is a rough patch of length $L = 12\ \mathrm{m}$ with a coefficient of kinetic friction $\mu_k = 0.7$.

If point D is just beyond the rough section, will the block be able to reach it? If not, how far through the rough patch can it travel?

The total mechanical energy at points A, B, and C are all equal. We know the block can reach point D if the friction requires a longer distance than $12\ \mathrm{m}$ for the kinetic energy by point C to be totally dissipated.

We can start by finding the velocity at point C because the gravitational potential energy of the block is constant between point C and point D.

$$\begin{aligned}
E_A &= E_C \\
\frac{1}{2}mv_A^2 + mgh_1 &= \frac{1}{2}mv_C^2 + mgh_2 \\
\frac{1}{2}v_A^2 + gh_1 &= \frac{1}{2}v_C^2 + gh_2 \\
\frac{1}{2}v_A^2 + g(h_1 - h_2) &= \frac{1}{2}v_C^2 \\
v_A^2 + 2g(h_1 - h_2) &= v_C^2 \\
v_C^2 &= (7\ \mathrm{m/s})^2 + 2(9.81\ \mathrm{m/s^2})(6\ \mathrm{m} - 2\ \mathrm{m}) \\
v_C^2 &= 127.48\ \mathrm{m^2/s^2}
\end{aligned}$$

Now, we can equate the kinetic energy at point C to the work done by friction and solve for the distance needed to fully dissipate it.

$$\begin{aligned}
K_C &= W_f \\
\frac{1}{2}mv_C^2 &= \mu_k mgd \\
d &= \frac{v_C^2}{2\mu_k g} \\
d &= \frac{127.48\ \mathrm{m^2/s^2}}{2(0.7)(9.81\ \mathrm{m/s^2})} \\
d &= \frac{127.48\ \mathrm{m^2/s^2}}{2(0.7)(9.81\ \mathrm{m/s^2})} \\
d &= 9.28\ \mathrm{m}
\end{aligned}$$

Thus, the block cannot reach point D, and will instead travel $9.28\ \mathrm{m}$ across the rough patch.

7.4 Force and Potential Energy

- Recall that the work done by a conservative force equals the negative change in potential energy.

$$W = -\Delta U$$

- Thus, given force $\vec{F}$ and potential energy U:

$$\vec{F} = -\vec{\nabla} U = -\frac{\partial U}{\partial x}\hat{\imath} - \frac{\partial U}{\partial y}\hat{\jmath} - \frac{\partial U}{\partial z}\hat{k}$$

The "∂" symbol refers to taking a derivative with respect to one variable, holding everything else constant, known as a *partial derivative*. The nabla vector is known as the gradient vector, and is a vector consisting of all partial derivatives.

Example: Object's Potential Energy Function

A small object with mass $m = 0.0900$ kg moves along the $+x$-axis. The only force applied to the object is a conservative force with potential energy function $U(x) = -\alpha x^2 + \beta x^3$ where $\alpha = 9.50\ \mathrm{J/m^2}$ and $\beta = 0.300\ \mathrm{J/m^3}$. The object is released from rest at negligible x.

1. When the object is at $x = 4.00$ m, what is its speed?
2. When the object is at $x = 4.00$ m, what is the magnitude of its acceleration?
3. What is the maximum value of x reached by the object during its motion?

---

1) Because we know the object starts from rest, the work done by the conservative force must be equal to the kinetic energy. Thus, we can equate kinetic energy to the work done, or the negative potential energy function (because we start from a negligible x) and solve for the velocity.

$$\begin{aligned}
K &= W \\
\frac{1}{2}mv^2 &= -U(4) \\
v &= \sqrt{-\frac{2U(4)}{m}} \\
v &= \sqrt{\frac{2(9.5\ \mathrm{J/m^2})(4\ \mathrm{m})^2}{0.09\ \mathrm{kg}} - \frac{2(0.3\ \mathrm{J/m^3})(4\mathrm{m}^3)}{0.09\ \mathrm{kg}}} \\
\boldsymbol{v} &= \mathbf{54.3\ m/s}
\end{aligned}$$

2) We can start by finding the negative derivative to get the the force function, then use the force function to get the acceleration function.

$$\begin{aligned}
F &= -\frac{\mathrm{d}U}{\mathrm{d}x} \\
F &= 2\alpha x - 3\beta x^2 = ma \\
a &= \frac{2\alpha x}{m} - \frac{3\beta x^2}{m} \\
a(4) &= \frac{2(9.5\ \mathrm{J/m^2})(4\ \mathrm{m})}{0.09\ \mathrm{kg}} - \frac{3(0.3\ \mathrm{J/m^3})(4\ \mathrm{m})^2}{0.09\ \mathrm{kg}} \\
\boldsymbol{a}(4) &= \mathbf{684\ m/s^2}
\end{aligned}$$

3) Similarly to our logic for part one, we can use the fact that $K = W = -U(x)$. By finding the zeros of $-U(x)$, we find that $x_{\max}$ is whenever the kinetic energy is zero again: $\boldsymbol{x_{\max}} = \mathbf{31.7\ m}$.

8 Momentum, Impulse, and Collisions

8.1 Momentum and Impulse

- **Momentum** is an object's *quantity of motion*, measured in $\mathrm{kg} \cdot \mathrm{m/s}$ or $\mathrm{N} \cdot \mathrm{s}$
  - Denoted p.

Definition: Momentum

Given momentum $\vec{p}$, mass m, and velocity $\vec{v}$

$$\vec{p} = m\vec{v}$$

- Thus, momentum is a vector quantity.
- We can also redefine Newton's second law in terms of momentum:

$$\Sigma\vec{F} = m\left(\frac{\mathrm{d}\vec{v}}{\mathrm{d}t}\right) = \frac{\mathrm{d}}{\mathrm{d}t}(m\vec{v}) = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t}$$

- An object's change in momentum is known as **impulse**.
  - Denoted J.

Definition: Impulse-Momentum Theorem

Given impulse $\vec{J}$ and momentum $\vec{p}$

$$\vec{J} = \Delta\vec{p}$$

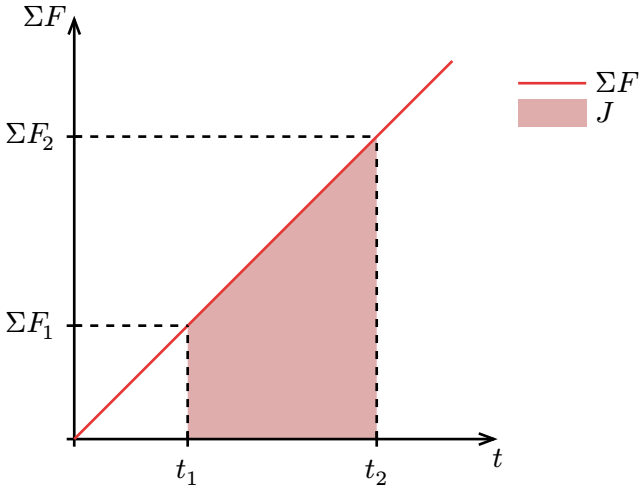
Derivation: Relating Impulse to Force

Given net force $\Sigma\vec{F}$, acceleration $\vec{a}$, and mass m

$$\begin{aligned}
\Sigma\vec{F} &= m\vec{a} \\
\Sigma\vec{F} &= m\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} \\
\Sigma\vec{F}\,\mathrm{d}t &= m\,\mathrm{d}\vec{v} \\
m\,\mathrm{d}\vec{v} &= \Sigma\vec{F}\,\mathrm{d}t \\
m\int_{v_1}^{v_2}\mathrm{d}\vec{v} &= \int_{t_1}^{t_2}\Sigma\vec{F}\,\mathrm{d}t \\
m\int_{v_1}^{v_2}\mathrm{d}\vec{v} &= \int_{t_1}^{t_2}\Sigma\vec{F}\,\mathrm{d}t \\
m(\vec{v}_2 - \vec{v}_1) &= \Sigma\vec{F}(t_2 - t_1) \\
m\Delta\vec{v} &= \Sigma\vec{F}\Delta t \\
\Delta\vec{p} &= \Sigma\vec{F}\Delta t \\
\vec{J} &= \Sigma\vec{F}\Delta t
\end{aligned}$$

If we interpret $\Sigma\vec{F}$ as a varying force, then impulse simply equals the integral of the net force with respect to time. If the impulse $\vec{J}$ and the time interval Δt are known, then $\Sigma\vec{F}$ represents the average net force over the time interval.

- It follows that impulse is the area under a force versus time graph.



- One real-life application of this concept is car crashes. The impulse experienced by a driver colliding with the steering wheel is the same as colliding with an air bag. However, because the air bag takes a longer time to stop the driver, the average force applied to the driver must be less.

Example: Ball Rebounding

You throw a 0.40 kg ball against a brick wall with a horizontal speed of 30 m/s. After colliding with the wall, the ball rebounds with a horizontal speed of 20 m/s.

1. Find the impulse of the net external force on the ball during the impact.
2. If the ball is in contact with the wall for 0.01 s, find the average horizontal force that the wall exerts on the ball during the impact.

---

1) Because we are given an initial and final velocity of the ball, we can apply the impulse-momentum theorem.

$$\begin{aligned}
J_x &= \Delta p_x \\
J_x &= mv_{2x} - mv_{1x} \\
J_x &= m(v_{2x} - v_{1x}) \\
J_x &= (0.4\ \text{kg})((-20\ \text{m/s}) - (30\ \text{m/s})) \\
\boldsymbol{J_x} &= \mathbf{-20\ N \cdot s}
\end{aligned}$$

2) Now that we know the impulse of the net external force, the average force can be calculated using the fact that the impulse is the area under a force versus time graph.

$$\begin{aligned}
J_x &= \Sigma F_{\text{av-}x} \Delta t \\
\Sigma F_{\text{av-}x} \Delta t &= J_x \\
\Sigma F_{\text{av-}x} &= \frac{J_x}{\Delta t} \\
\Sigma F_{\text{av-}x} &= \frac{20\ \text{N} \cdot \text{s}}{0.01\ \text{s}} \\
\boldsymbol{\Sigma F_{\text{av}-x}} &= \mathbf{2000\ N}
\end{aligned}$$

8.2 Conservation of Momentum

- **Internal forces** are the forces exerted by a system's particles on each other.
- **External forces** are the forces exerted by particles outside the system on the system's particles.
- **Isolated systems** are systems with no external forces.
- The **total momentum** of a system is the sum of the momenta of the system's particles.
  - Denoted P.

$$\vec{P} = \vec{p}_A + \vec{p}_B + \cdots = m\vec{v}_A + m\vec{v}_B + \cdots$$

Definition: Conservation of Momentum
We say that a system's total momentum is *conserved* if the system's net external force is zero.

- Note that by Newton's third law, a system's internal forces do not impact the system's total momentum.

Example: Gun Recoil
A marksman holds a 3.00 kg rifle loosely, allowing it to recoil. He fires a 5.00 g bullet whose horizontal velocity relative to the ground is 300 m/s.
1. What is the rifle's recoil velocity?
2. What are the final momentum and kinetic energy of the bullet and rifle?

---

1) During the shot, the external forces exerted by the marksman are negligible. Thus, the system's total momentum is conserved. Now, it follows that the system's total momentum is initially zero, so it should be zero after the shot. Since we are given everything but the rifle's recoil velocity, we can simply isolate it by equating the momenta to zero.

$$\begin{aligned} p_R + p_B &= 0 \\ p_R &= -p_B \\ m_R v_R &= -m_B v_B \\ v_R &= -\frac{m_B v_B}{m_R} \\ v_R &= -\frac{\left(5\text{ g} \cdot \frac{0.001\text{ kg}}{1\text{ g}}\right)(300\text{ m/s})}{3\text{ kg}} \\ \boldsymbol{v_R} &= \mathbf{-0.500\ m/s} \end{aligned}$$

2) Now that we have the recoil velocity of the rifle, we can simply plug our values into the momentum and kinetic energy formulas.

$$p_R = m_R v_R \qquad K_R = \frac{1}{2} m_R v_R^2 \qquad p_B = m_B v_B \qquad K_B = \frac{1}{2} m_B v_B^2$$

$$\boldsymbol{p_R} = \mathbf{-1.50\ kg \cdot m/s} \qquad \boldsymbol{K_R} = \mathbf{0.375\ J} \qquad \boldsymbol{p_B} = \mathbf{1.5\ kg \cdot m/s} \qquad \boldsymbol{K_B} = \mathbf{225\ J}$$

8.3 Momentum Conservation and Collisions

- Collisions are strong interactions between objects over a relatively short time interval.
  - Momentum is conserved in collisions, but kinetic energy is not always conserved.
- **Elastic collisions** occur if the forces between the colliding objects are conservative.

- Thus, the total kinetic energy is conserved.

$$\frac{1}{2}m_A v_{1A}^2 + \frac{1}{2}m_B v_{2A}^2 = \frac{1}{2}m_A v_{2A}^2 + \frac{1}{2}m_B v_{2B}^2$$

- **Inelastic collisions** occur if there are nonconservative forces between the colliding objects.
- **Completely inelastic collisions** are when colliding objects stick together and move as one object.
  - This leads to a unique formula for conservation of momentum in completely inelastic collisions:

$$m_A \vec{v}_A + m_B \vec{v}_B = (m_A + m_B)\vec{v}_{AB}$$

- In inelastic and completely inelastic collisions, kinetic energy is not conserved.

Completely Inelastic Collisions

Derivation: Loss of Kinetic Energy

In order to establish why a system's kinetic energy decreases in completely inelastic collisions, we can first find the velocity after the inelastic collision using conservation of momentum. To simplify it further, we can say that object B is initially at rest.

$$\begin{aligned}(m_A + m_B)\vec{v}_{AB} &= m_A \vec{v}_A + m_B \vec{v}_B \\ (m_A + m_B)\vec{v}_{AB} &= m_A \vec{v}_A \\ \vec{v}_{AB} &= \frac{m_A \vec{v}_A}{m_A + m_B}\end{aligned}$$

Now, we can use their magnitudes to find initial and final kinetic energies of the system in terms of object A's original velocity and find the ratio between them.

$$\begin{aligned}K_1 &= \frac{1}{2}m_A v_A^2 \\ K_2 &= \frac{1}{2}(m_A + m_B)v_{AB}^2 \\ &= \frac{1}{2}(m_A + m_B)\left(\frac{m_A v_A}{m_A + m_B}\right)^2 \\ &= \frac{m_A^2 v_A^2}{2(m_A + m_B)} \\ \therefore \frac{K_2}{K_1} &= \frac{m_A}{m_A + m_B}\end{aligned}$$

Now, it follows that this ratio is always less than one, showing that kinetic energy is always lost in completely inelastic collisions. This holds true even if object B is not originally at rest.

- Just because a collision is inelastic *does not mean* that the colliding objects have to stick together. For instance, kinetic energy could be dissipated from deformations.

Elastic Collisions

- Commonly, elastic collision problems start with one object at rest before the collision.
- Thus, we can establish a relationship between the initial velocity of the moving object and the final velocity of the object initially at rest.

Derivation: When a Moving Object Collides with a Still Object

To establish a relationship between the initial velocity of the moving object and the final velocity of the object initially at rest, we can apply the conservation of energy and momentum.

$$\frac{1}{2}m_A v_{1A}^2 = \frac{1}{2}m_A v_{2A}^2 + \frac{1}{2}m_B v_B^2 \qquad m_A v_{1A} = m_A v_{2A} + m_B v_B$$

$$\frac{1}{2}m_A(v_{1A}^2 - v_{2A}^2) = \frac{1}{2}m_B v_B^2 \qquad m_A(v_{1A} - v_{2A}) = m_B v_B$$

$$\frac{1}{2}m_A(v_{1A} + v_{2A})(v_{1A} - v_{2A}) = \frac{1}{2}m_B v_B^2$$

Now, by dividing the kinetic energy equation by the momentum equation, we can derive the final velocity of object B in terms of object A's velocities.

$$\frac{1}{2}v_{1A} + \frac{1}{2}v_{2A} = \frac{1}{2}v_B$$
$$v_B = v_{1A} + v_{2A}$$

Using this definition, we can define the final velocity of object A (which we are not interested in) in terms of object A's initial velocity using the momentum equation.

$$\begin{aligned} m_A v_{1A} &= m_A v_{2A} + m_B(v_{1A} + v_{2A}) \\ m_A v_{1A} &= m_A v_{2A} + m_B v_{1A} + m_B v_{2A} \\ m_A v_{1A} - m_B v_{1A} &= m_A v_{2A} + m_B v_{2A} \\ v_{2A} &= \frac{m_A v_{1A} - m_B v_{1A}}{m_A + m_B} \\ v_{2A} &= \frac{m_A - m_B}{m_A + m_B} v_{1A} \end{aligned}$$

We can plug this value for the final velocity of the moving object back into the velocity relationship we defined earlier to establish the relationship we are looking for.

$$\begin{aligned} v_B &= v_{1A} + \frac{m_A - m_B}{m_A + m_B} v_{1A} \\ v_B &= \frac{m_A + m_B}{m_A + m_B} v_{1A} + \frac{m_A - m_B}{m_A + m_B} v_{1A} \\ v_B &= \frac{2m_A}{m_A + m_B} v_{1A} \end{aligned}$$

We can interpret this answer in numerous ways.

- Say that the mass of the moving object is much less than the still object. Then, the still object's velocity after the collision is very small.
  - If a ping pong ball collides with a still cannon ball, the cannon ball shouldn't move much.
- If the mass of the moving object is equal to the still object, then the still object's velocity after the collision will be equal to the moving object's initial velocity.
  - In a game of pool, a cue ball of roughly equal mass to the other balls is used to push the other balls.
- If the mass of the moving object is much greater than the still object, the still object's velocity after the collision is very high.
  - If a cannon ball collides with a still ping pong ball, the ping pong ball should move a lot.

- Now, recall the the equality we determined in the previous derivation:

$$v_B = v_{1A} + v_{2A}$$

- If we rearrange this equality, we can get the *velocity of the object B relative to object A* because these velocities are given relative to the ground:

$$v_{1A} = v_B - v_{2A}$$

8.4 Center of Mass

- A system's **center of mass** is the average position of all matter in the system.

Definition: Center of Mass with Point Masses

Given point masses m_i of a system each a distance x_i, y_i, or z_i from the origin, center of masses x_{cm}, y_{cm}, and z_{cm} are defined as follows:

$$x_{\text{cm}} = \frac{\sum_{i=1}^{N} m_i x_i}{\sum_{i=1}^{N} m_i} \qquad y_{\text{cm}} = \frac{\sum_{i=1}^{N} m_i y_i}{\sum_{i=1}^{N} m_i} \qquad z_{\text{cm}} = \frac{\sum_{i=1}^{N} m_i z_i}{\sum_{i=1}^{N} m_i}$$

Or, condensed into a single vector equation:

$$\vec{R}_{\text{cm}} = \frac{\sum_{i=1}^{N} m_i \vec{r}}{\sum_{i=1}^{N} m_i}$$

Definition: Center of Mass with Continuous Mass Distribution

Given distance $\vec{r}$ from the origin and total system mass M_{tot}, center of mass $\vec{R}_{\text{cm}}$ is defined as follows:

$$\vec{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}} \int \vec{r} \cdot \mathrm{d}m$$

- Because point masses are not given, we integrate with respect to the mass throughout the object.
- To apply this definition, we can find the differential in terms of densities and also integrate over different axes depending on the dimensions involved in the mass distribution.

Linear density, denoted λ:

$$\lambda = \frac{M}{L} = \frac{\mathrm{d}m}{\mathrm{d}L}$$
$$\therefore \mathrm{d}m = \lambda\,\mathrm{d}L = \lambda\,\mathrm{d}x$$

Surface density, denoted σ:

$$\sigma = \frac{M}{A} = \frac{M}{L^2} = \frac{\mathrm{d}m}{\mathrm{d}A}$$
$$\therefore \mathrm{d}m = \sigma\,\mathrm{d}A = \sigma\,\mathrm{d}x\,\mathrm{d}y$$

Volume density, denoted ρ:

$$\rho = \frac{M}{V} = \frac{M}{L^3} = \frac{\mathrm{d}m}{\mathrm{d}V}$$
$$\therefore \mathrm{d}m = \lambda\,\mathrm{d}V = \rho\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$

Derivation: Center of Mass of a Uniform Rod

Intuitively, the center of mass of a uniform rod should be the middle. We can use the center of mass equation for continuous mass distributions to prove this.

Continued on next page

Derivation: Center of Mass of a Uniform Rod *continued*

We can define the length of any rod as L along the x-axis, and define $\mathrm{d}m$ using linear density λ.

$$x_{\text{cm}} = \frac{1}{M_{\text{tot}}} \int x \, \mathrm{d}m$$

$$x_{\text{cm}} = \frac{1}{M_{\text{tot}}} \int x\lambda \, \mathrm{d}L$$

$$x_{\text{cm}} = \frac{\lambda}{M_{\text{tot}}} \int_0^L x \, \mathrm{d}x$$

$$x_{\text{cm}} = \frac{\lambda}{M_{\text{tot}}} \left(\frac{1}{2}x^2\right)\Big|_0^L$$

$$x_{\text{cm}} = \frac{\lambda}{M_{\text{tot}}} \cdot \frac{L^2}{2}$$

$$x_{\text{cm}} = \frac{M_{\text{tot}}}{L} \cdot \frac{1}{M_{\text{tot}}} \cdot \frac{L^2}{2}$$

$$x_{\text{cm}} = \frac{1}{L} \cdot \frac{L^2}{2}$$

$$x_{\text{cm}} = \frac{L}{2}$$

Derivation: Center of Mass of a Uniform Cube

Similar to a uniform rod, the center of mass of a uniform cube is at the very center. Again, we can use the center of mass equation for continuous mass distributions to prove this.

We can define the side lengths of any cube as L and define $\mathrm{d}m$ using volume density ρ.

$$\vec{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}} \int \vec{r} \cdot \mathrm{d}m$$

$$\vec{R}_{\text{cm}} = \frac{1}{M_{\text{tot}}} \iiint \vec{r} \cdot \rho \, \mathrm{d}V$$

$$\vec{R}_{\text{cm}} = \frac{\rho}{M_{\text{tot}}} \iiint \vec{r} \cdot \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$$

$$\vec{R}_{\text{cm}} = \frac{\rho}{M_{\text{tot}}} \int_0^L \int_0^L \int_0^L \left(x\hat{\imath} + y\hat{\jmath} + z\hat{k}\right) \cdot \mathrm{d}x \, \mathrm{d}y \, \mathrm{d}z$$

$$\vec{R}_{\text{cm}} = \frac{\rho}{M_{\text{tot}}} \int_0^L \int_0^L \left(\frac{1}{2}x^2\hat{\imath} + xy\hat{\jmath} + xz\hat{k}\right)\Big|_{x=0}^{x=L} \cdot \mathrm{d}y \, \mathrm{d}z$$

$$\vec{R}_{\text{cm}} = \frac{\rho}{M_{\text{tot}}} \int_0^L \int_0^L \left(\frac{L^2}{2}\hat{\imath} + Ly\hat{\jmath} + Lz\hat{k}\right) \cdot \mathrm{d}y \, \mathrm{d}z$$

$$\vec{R}_{\text{cm}} = \frac{\rho}{M_{\text{tot}}} \int_0^L \left(\frac{L^2}{2}y\hat{\imath} + \frac{L}{2}y^2\hat{\jmath} + Lyz\hat{k}\right)\Big|_{y=0}^{y=L} \cdot \mathrm{d}z$$

Continued on next page

Derivation: Center of Mass of a Uniform Cube *continued*

$$\vec{R}_{\text{cm}} = \frac{\rho}{M_{\text{tot}}} \int_0^L \left(\frac{L^3}{2}\hat{\imath} + \frac{L^3}{2}\hat{\jmath} + L^2 z\hat{k} \right) \cdot \mathrm{d}z$$

$$\vec{R}_{\text{cm}} = \frac{\rho}{M_{\text{tot}}} \left(\frac{L^3}{2}z\hat{\imath} + \frac{L^3}{2}z\hat{\jmath} + \frac{L^2}{2}z^2\hat{k} \right) \Bigg|_{z=0}^{z=L}$$

$$\vec{R}_{\text{cm}} = \frac{\rho}{M_{\text{tot}}} \left(\frac{L^4}{2}\hat{\imath} + \frac{L^4}{2}\hat{\jmath} + \frac{L^4}{2}\hat{k} \right)$$

$$\vec{R}_{\text{cm}} = \frac{M_{\text{tot}}}{V} \cdot \frac{1}{M_{\text{tot}}} \left(\frac{L^4}{2}\hat{\imath} + \frac{L^4}{2}\hat{\jmath} + \frac{L^4}{2}\hat{k} \right)$$

$$\vec{R}_{\text{cm}} = \frac{1}{V} \cdot \left(\frac{L^4}{2}\hat{\imath} + \frac{L^4}{2}\hat{\jmath} + \frac{L^4}{2}\hat{k} \right)$$

$$\vec{R}_{\text{cm}} = \frac{1}{L^3} \cdot \left(\frac{L^4}{2}\hat{\imath} + \frac{L^4}{2}\hat{\jmath} + \frac{L^4}{2}\hat{k} \right)$$

$$\vec{R}_{\text{cm}} = \frac{L}{2}\hat{\imath} + \frac{L}{2}\hat{\jmath} + \frac{L}{2}\hat{k}$$

Motion of the Center of Mass

- Like we've previously established, we can find the velocity and acceleration of the center of mass using derivatives.

Definition: Velocity of the Center of Mass

Given point masses m_i of a system each moving with velocity $\vec{v}_i$, the velocity of the center of mass $\vec{v}_{\text{cm}}$ is defined as follows:

$$\vec{v}_{\text{cm}} = \frac{\sum_{i=1}^{N} m_i \vec{v}_i}{\sum_{i=1}^{N} m_i}$$

- Now, say the total mass of the system is $\sum_{i=1}^{N} m_i = M_{\text{tot}}$. If we multiply the velocity of the center of mass by the total mass, we arrive at a new definition for total momentum $\vec{P}$:

$$M_{\text{tot}}\vec{v}_{\text{cm}} = \sum_{i=1}^{N} m_i \vec{v}_i = m_1\vec{v}_1 + m_2\vec{v}_2 + \cdots + m_N\vec{v}_N = \vec{P}$$

- Thus, a system with a net external force of zero has a center of mass that moves with constant velocity.

Definition: Acceleration of the Center of Mass

Given point masses m_i of a system each with acceleration $\vec{a}_i$, the acceleration of the center of mass $\vec{a}_{\text{cm}}$ is defined as follows:

$$\vec{a}_{\text{cm}} = \frac{\sum_{i=1}^{N} m_i \vec{a}_i}{\sum_{i=1}^{N} m_i}$$

- Now, it follows that multiplying the acceleration of the center of mass by the system's total mass yields the rate of change of the system's total momentum: the *net force.*

$$M_{\text{tot}} \vec{a}_{\text{cm}} = \sum_{i=1}^{N} m_i \vec{a}_i = m_1 \vec{a}_1 + m_2 \vec{a}_2 + \cdots + m_N \vec{a}_N = \Sigma \vec{F}$$

- However, due to Newton's third law, the net internal forces cancel out, leaving just the net external forces. Thus:

$$M_{\text{tot}} \vec{a}_{\text{cm}} = \Sigma \vec{F} = \Sigma \vec{F}_{\text{ext}} + \Sigma \vec{F}_{\text{int}} = \Sigma \vec{F}_{\text{ext}}$$

- Now, imagine a scenario where a shell is shot into the air, following a parabolic trajectory. Even when it splits into multiple fragments that each follow their own trajectory, the center of mass *follows the original trajectory* as if the masses of the system were all concentrated on that path.
- This explains the definition of net force as shown in chapter 4.

9 Rotation of Rigid Bodies

10 Dynamics of Rotational Motion

11 Equilibrium and Elasticity

14 Periodic Motion